

Reading

PREPARING TO READ

A BUILD VOCABULARY The words in **blue** are used in the reading passage. Match each word to its definition. Use a dictionary to help you.

- | | |
|-------------------------------|--|
| 1. ____ adaptation (n) | a. to look similar to something else |
| 2. ____ arise (v) | b. the people employed in an organization or place |
| 3. ____ distinct (adj) | c. bad luck |
| 4. ____ dominant (adj) | d. to start to develop or exist |
| 5. ____ misfortune (n) | e. the act of changing in order to survive in a particular environment |
| 6. ____ resemble (v) | f. more powerful than other groups |
| 7. ____ workforce (n) | g. noticeable or different |

B BUILD VOCABULARY Complete the sentences below with the correct form of the words in **blue**. Use a dictionary to help you.

associate (v)
radically (adv)

burden (n)
sole (adj)

corruption (n)
thrive (v)

mining (n)
undergo (v)

1. The country _____ a transformation in the 1970s, after it signed the trade deal.
2. They grew up there, so they _____ that place with their youth.
3. After she announced the bad news, the mood in the room changed _____.
4. There are a lot of minerals in the ground, so _____ is a big industry here.
5. Creating more jobs for the unemployed is the _____ purpose of this proposal.
6. Her ambition is to work hard and _____ in the city.
7. The cost of housing is a huge _____ for people living in the city.
8. If we want to reduce crime, we need to fix _____ in the police force.

C USE VOCABULARY Discuss these questions with a partner.

1. What are some changes your country has **undergone** in the last 20 years?
2. What do people usually **associate** your country or region with?

Critical Thinking |

D PREDICT Read the first two paragraphs of the reading passage. How do you think Africa's shape affects life on the continent? Discuss with a partner.

DEVELOPING READING SKILLS

READING SKILL Annotating a Text

Annotating—or marking up—a text while you read it helps you to stay focused on what you are reading. It also helps you remember and find important information later. Here are some ways to annotate a text:

- Highlight important ideas. Use one color for main ideas and another for supporting information.
- Circle new vocabulary to check later.
- Underline and put a question mark next to parts you don't understand, to re-read later.
- Break the text into sections by sub-topic and label each section in the margins.

Here is an example of a paragraph with annotations:

For people in developing nations, solar-powered devices can offer several distinct advantages. The availability of low-cost solar lamps, for example, means longer working hours and better security at night. One organization in Uganda—Solar Sister—is using solar power to turn local women into solar entrepreneurs. It employs them to sell its solar-powered products. The women don't require qualifications or experience to start because Solar Sister provides them with training. Furthermore, Solar Sister empowers these women to start businesses of their own by providing them with products to sell at no initial financial cost. ?

In the paragraph above, a reader has highlighted the main idea in pink and supporting ideas in blue. An unfamiliar word, *entrepreneurs*, is circled. A question mark indicates that the student is unsure about the last underlined sentence and plans to re-read it.

A ANNOTATE A TEXT The paragraphs below are from the reading passage. Annotate them using the methods in the Reading Skill box above.

Ask someone to tell you quickly what they associate with Africa and the answers you'll get will probably range from "cradle of humankind" and "big animals" to "poverty" and "tribalism." How did one continent come to embody such extremes?

Geography and history go a long way toward providing the explanations. Geographically, Africa resembles a bulging sandwich. The sole continent to span both the north and south temperate zones, it has a thick tropical core lying between one thin temperate zone in the north and another in the south. That simple geographic reality explains a great deal about Africa today.

B DISCUSS Work with a partner. Compare your annotations. Are there any other ways you could annotate these paragraphs? Discuss your ideas.

C ANNOTATE A TEXT Read the entire reading passage. Annotate as you read.

THE SHAPE OF AFRICA

By Jared Diamond



The hope for Africa's future lies with its abundant human and natural resources.

A  Ask someone to tell you quickly what they **associate** with Africa and the answers you'll get will probably range from "cradle of humankind" and "big animals" to "poverty" and "tribalism." How did one continent come to embody such extremes?

B Geography and history go a long way toward providing the explanations. Geographically, Africa **resembles** a bulging sandwich. The **sole** continent to span both the north and south temperate zones,¹ it has a thick tropical core lying between one thin temperate zone in the north and another in the south. That simple geographic reality explains a great deal about Africa today.

C As to its human history, this is the place where—some seven million years ago—the evolutionary lines of apes and protohumans² diverged. It remained the only continent our ancestors inhabited until around two million years ago, when *Homo erectus* expanded out of Africa into Europe and Asia. Over the next 1.5 million years, the populations of those three continents followed such different evolutionary courses that they became **distinct** species. Europe's became the Neanderthals, Asia's remained *Homo erectus*, but Africa's evolved into our own species, *Homo sapiens*. Sometime between 100,000 and 50,000 years ago, our African ancestors **underwent** some further profound change. Whether it was the development of complex speech or

something else, such as a change in brain wiring, we aren't sure. Whatever it was, it transformed those early *Homo sapiens* into what paleoanthropologists³ call "behaviorally modern" *Homo sapiens*. Those people, probably with brains similar to our own, expanded again into Europe and Asia. Once there, they exterminated or replaced or interbred with Neanderthals and Asia's hominins⁴ and became the **dominant** human species throughout the world.

D In effect, Africans enjoyed not just one but three huge head starts over humans on other continents. That makes Africa's economic struggles today, compared with the successes of other continents, particularly puzzling. It's the opposite of what one would expect from the runner first off the block. Here again, geography and history give us answers.

E It turns out that the rules of the competitive race among the world's humans changed **radically** about 10,000 years ago, with the origins of agriculture. The domestication of wild plants and animals meant our ancestors could grow their own food instead of having to hunt or gather it in the wild. That allowed people to settle in permanent villages, to increase their populations, and to feed specialists—inventors, soldiers, and kings—who did not produce food. With domestication came other advances, including the first metal tools, writing, and state societies.

¹ Temperate zones are areas between the tropics and the polar circle.

² A protohuman is an early ancestor of modern humans.

³ Paleoanthropologists are scientists who study human fossils.

⁴ Hominins are the early forms of humans that descended from primates.

Africa's unique geography makes agriculture in much of the continent challenging.



F The problem is that only a tiny minority of wild plants and animals lend themselves to domestication, and those few are concentrated in about half a dozen parts of the world. As every schoolchild learns, the world's earliest and most productive farming arose in the Fertile Crescent of southwestern Asia, where wheat, barley, sheep, cattle, and goats were domesticated. While those plants and animals spread east and west in Eurasia, in Africa they were stopped by the continent's north-south orientation. Crops and livestock tend to spread much more slowly from north to south than from east to west because different latitudes require **adaptation** to different climates, seasonalities, day lengths, and diseases. Africa's own native plant species—sorghum, oil palm, coffee, millets, and yams—weren't domesticated until thousands of years after Asia and Europe had agriculture. And Africa's

geography kept oil palm, yams, and other crops of equatorial Africa from spreading into southern Africa's temperate zone. While South Africa today boasts⁵ the continent's richest agricultural lands, the crops grown there are mostly northern temperate crops, such as wheat and grapes, brought directly on ships by European colonists. Those same crops never succeeded in spreading south through the thick tropical core of Africa.

G The domesticated sheep and cattle of Fertile Crescent origins took about 5,000 years to spread from the Mediterranean down to the southern tip of Africa. The continent's own native animals—with the exception of guinea fowl and possibly donkeys and one breed of cattle—proved impossible to domesticate. History might have turned out differently if African armies, fed by barnyard-giraffe meat and backed by waves of cavalry⁶ mounted on huge rhinos, had swept into Europe to

⁵To boast something is to possess a feature that is a source of pride.

⁶A cavalry is a group of soldiers who ride horses.

overrun its mutton-fed soldiers mounted on puny horses. That this didn't happen was no fault of the Africans; it was because of the kinds of wild animals available to them.

H Ironically, the long human presence in Africa is probably the reason the continent's species of big animals survive today. African animals coevolved with humans for millions of years, as human hunting prowess gradually progressed from the basic skills of our early ancestors. That gave the animals time to learn a healthy fear of man and, with it, a healthy avoidance of human hunters. In contrast, North and South America and Australia were settled by humans only within the last tens of thousands of years. To the **misfortune** of the big animals of those continents, the first humans they encountered were already fully modern people, with modern brains and hunting skills. Most of those animals—woolly mammoths, saber-toothed cats, and, in Australia, marsupials⁷ as big as rhinoceroses—

disappeared soon after humans arrived. Entire species may have been exterminated before they had time to learn to be wary of hunters.

I Unfortunately, the long human presence in Africa also encouraged something else to **thrive**—diseases. The continent has a well-deserved reputation for having spawned some of our nastiest ones: malaria, yellow fever, East African sleeping sickness, and AIDS. These and many other human illnesses arose when microbes causing disease in animals crossed species lines to evolve into a human disease. For a microbe already adapted to one species, to adapt to another can be difficult and require a lot of evolutionary time. Much more time has been available in Africa, cradle of humankind, than in any other part of the planet. That's half the answer to Africa's disease **burden**; the other half is that the animal species most closely related to humans—those whose microbes required the least adaptation to jump species—are the African great apes and monkeys.

⁷Marsupials are animals such as kangaroos. Female marsupials carry their babies in pouches on their bellies.

Like this oryx antelope, much of Africa's wildlife can't easily be domesticated.



J Africa continues to be shaped in other ways by its long history and its geography. Of mainland Africa's ten richest countries, eight lie partly or entirely within its temperate zones: Egypt, Libya, Tunisia, and Algeria in the north; and Angola, South Africa, Botswana, and Namibia in the south. Gabon and Equatorial Guinea are Africa's only tropical countries to make the list. In addition, nearly a third of the countries of mainland Africa (15 out of 47) are landlocked, and the only African river navigable from the ocean for long distances inland is the Nile. Since waterways provide the cheapest way to transport cumbersome⁸ goods, geography again thwarts Africa's progress.

K All these factors can lead to the question: "Is the continent, or at least its

big tropical core, doomed eternally to wars, poverty, and devastating diseases?" I'd answer, "Absolutely not." On my own visits to Africa, I've been struck by how harmoniously ethnic groups live together in many countries—far better than they do in many other parts of the globe. Tensions **arise** in Africa, as they do elsewhere, when people see no other way out of poverty except to fight their neighbors for dwindling resources. But many areas of Africa have an abundance of resources: The rivers of central Africa are great generators of hydroelectric power; the big animals are a major source of ecotourism revenue in eastern and southern Africa; and the forests in the

⁸If something is cumbersome, it is large and heavy and therefore difficult to carry or handle.



wetter regions, if managed and logged sustainably, would be renewable and lucrative⁹ sources of income.

- L As for Africa's health problems, they can be greatly alleviated with the right planning and funding. Within the past half century, several formerly poor countries in Asia recognized that tropical diseases were a major drain on their economies. By investing in public health measures, they have successfully curbed those diseases, and the increased health of their people has led to far healthier economies. Within Africa itself, some international **mining** and oil companies have been funding successful public health programs throughout their concession areas¹⁰ because they

realized that protecting the health of their workers was an excellent business investment for them.

- M What's the best case for Africa's future? If the continent can overcome its health problems and the **corruption** that plagues many of its governments and institutions, then it could take advantage of today's globalized, technological world in much the same way that China and India are now doing. Technology could give Africa the connections that its geography, particularly its rivers, long denied it. Nearly half of all African countries are English speaking—an advantage in trade relations—and an educated, English-speaking **workforce** could well attract service jobs to many African countries.

- N If Africa is to head into a bright future, outside investment will continue to be needed, at least for a time. The cost of perpetual aid to or military intervention in Africa is thousands of times more expensive than solving its health problems and supporting local development, thereby heading off¹¹ conflicts. The entire world will be healthier and safer if Africa's nations increasingly take their places as peaceful and prospering members of the world community.

Adapted from "The Shape of Africa," by Jared Diamond: National Geographic Magazine, September 2005

Jared Diamond is an American ecologist, geographer, and anthropologist. His book *Guns, Germs, and Steel: The Fates of Human Societies* won the Pulitzer Prize.

⁹ A lucrative activity, job, or business is one that's very profitable.

¹⁰ A concession area is a place where someone is given the right to sell a product or run a business.

¹¹ If you head off an event, you prevent it from happening.



A vibrant, thriving community in Kinshasa, Democratic Republic of the Congo

UNDERSTANDING THE READING

A UNDERSTAND MAIN IDEAS Match paragraphs B–H to their main ideas.

1. _____ Paragraph B a. African animals are resilient because they evolved alongside humans.
2. _____ Paragraph C b. Humans have been in Africa for a very long time.
3. _____ Paragraph D c. Africa's geographical orientation restricted the spread of agriculture.
4. _____ Paragraph E d. Despite Africa's long human history, much of the continent has not thrived.
5. _____ Paragraph F e. Africa's geography explains the extremes that exist on the continent.
6. _____ Paragraph G f. The development of agriculture impacts a culture.
7. _____ Paragraph H g. Most of the animals in Africa were unsuitable for domestication.

Review
Paraphrasing
in Unit 5

B UNDERSTAND MAIN IDEAS Write the main ideas of the paragraphs below.

1. Paragraph I: The long human presence in Africa led to _____

2. Paragraph K: Africa has hope because _____

3. Paragraph L: _____

4. Paragraph M: _____

Critical Thinking

C UNDERSTAND CHRONOLOGY Look again at paragraph C. Put the events (a–f) in the correct order in the timeline.

- a. "Behaviorally modern" Homo sapiens appears in Africa.
- b. Protohumans remain as Homo erectus in Asia, but evolve into Neanderthals in Europe and Homo sapiens in Africa.
- c. Apes and protohumans split.
- d. Homo erectus moves to Europe and Asia.
- e. Homo sapiens becomes the dominant species as other forms of humans die out.
- f. Homo sapiens moves into Europe and Asia.

